Visual Explainer Cards: Turning Complex Papers into Accessible Knowledge

The rapid growth of scientific publishing has created a widening gap between specialized research and public understanding. Each year, over 3 million peer-reviewed papers are published, yet fewer than 0.1% ever reach a non-specialist audience. This communication bottleneck prevents valuable insights from influencing policy, education, and public discourse.

Traditional science communication relies on human mediators?journalists, educators, and popularizers?to translate dense academic prose into accessible formats. However, this pipeline cannot scale to match the volume of published research. Even well-funded science communication initiatives cover only a tiny fraction of the literature.

We identify three core challenges: (1) the exponential growth of scientific output outpaces human translation capacity; (2) existing automated summarization tools produce text-only outputs that fail to leverage the pedagogical power of visual explanation; and (3) most readers abandon complex papers within the first two pages, never reaching the insights buried in methods and results sections.

This paper proposes Visual Explainer Cards (VEC)?a system that automatically transforms academic papers into structured, illustrated explainer cards. Each card targets a single concept, pairing a plain-language explanation with a visual metaphor, key takeaways, and source references. The system is designed to operate locally, without requiring external AI APIs, making it suitable for privacy-sensitive domains such as medical research and legal analysis.

Proposed Approach: The 8-Step Teaching Pipeline

Our approach is grounded in two observations from cognitive science: (1) the generation effect shows that restructuring information into new formats improves comprehension, and (2) dual coding theory demonstrates that combining verbal and visual channels enhances learning.

The VEC pipeline consists of eight sequential stages, each inspired by established pedagogical principles:

Stage 1 ? Problem Identification: Extract the core research question from the introduction. We use heuristic keyword matching to locate problem statements, gap declarations, and motivation clauses.

Stage 2 ? Significance Framing: Identify why the problem matters. We scan for phrases indicating real-world impact, economic implications, or societal relevance.

Stage 3 ? Baseline Comparison: Extract descriptions of prior approaches. We leverage section heading heuristics and transition phrases like 'previous work', 'existing methods', and 'state of the art'.

Stage 4 ? Conceptual Mapping: Build a concept graph from key terms. We extract noun phrases and named entities, then construct a co-occurrence network weighted by paragraph proximity.

Stage 5 ? Mechanism Decomposition: Break down the proposed method into sequential steps. We detect enumerations (first, second, then), process descriptions (the system computes, the algorithm selects), and data flows.

Stage 6 ? Evidence Compilation: Aggregate findings, statistics, and performance metrics. We extract sentences containing numerical comparisons, percentage improvements, and statistical significance markers.

Stage 7 ? Limitation Enumeration: Identify caveats, assumptions, boundary conditions, and acknowledged weaknesses. We scan for hedging language, limitation sections, and future work declarations.

Stage 8 ? Impact Synthesis: Compose a unified takeaway that connects the problem, solution, and implications into a concise narrative arc.

Evaluation, Limitations, and Future Directions

We evaluated VEC on a corpus of 500 open-access papers spanning computer science, biomedicine, and social sciences. Human raters (n=12, all graduate students with domain expertise) assessed the generated cards on three dimensions: factual accuracy (mean 4.2/5.0), pedagogical clarity (mean 4.0/5.0), and visual metaphor appropriateness (mean 3.7/5.0).

Inter-rater reliability was moderate (Fleiss kappa = 0.62), suggesting consistent but not uniform judgment. The lowest-scoring cards typically involved highly technical methods sections where heuristic extraction struggled to identify the core mechanism.

Current limitations include: (1) The system cannot parse figures, tables, or mathematical notation embedded in PDFs?these are simply omitted from text extraction. (2) Multi-column academic layouts occasionally produce out-of-order text, confusing the section detection heuristics. (3) The visual metaphors are drawn from a fixed palette of eight geometric templates, which may not suit all subject domains. (4) Scanned PDFs without embedded text layers are not supported, as we intentionally exclude OCR from the pipeline to maintain simplicity and speed.

Several promising directions remain: integrating lightweight language models for more nuanced concept extraction, adding support for figure and table parsing, and developing domain-specific visual metaphor libraries for fields like chemistry, physics, and medicine.

The source code is available as open source under the MIT license. We invite contributions from the community to extend the pipeline and improve extraction quality across diverse document formats.